

Irene Aldridge

Let's Use AI to Improve US Education I commend President Trump and his team for taking the leadership stance on Artificial Intelligence and am grateful for the opportunity to comment on President Trump's initiatives. I propose to use AI to help U.S. teachers and students improve their Mathematics and Science education. Artificial intelligence (AI) is a vast topic that includes algorithmic information processing, prediction, human-computer interaction, and much more. As a graduate level instructor who has taught machine learning (ML) and AI at Cornell University Operations Research department (ORIE) Masters' level programs for the past 8 years, I have continuously expanded my knowledge to stay at the cusp of the innovation in the space to add maximum value to the students. As such, I think I can competently comment on the importance of AI in the years to come. I regularly teach Chinese students in my classes at Cornell and have the opportunity to observe just how far ahead their knowledge of Mathematics is compared to that of their U.S. peers. The Chinese students are, on average, about 10 study years ahead of the US students. The gap is primarily driven by the higher pace and more advanced content taught in the schools in China. As a mother of 3 school-age children, I have had first-hand experience with the US education system and strongly believe that AI can help us close the gap. Specifically, AI is able to provide individualized curriculum and attention to students at scale that has never been previously imagined or possible. Artificial Intelligence is unstoppable. It is not a fad, a flash in the pan, or a blip on a radar. It is an innovation made possible by the increasingly cheaper and more powerful computing technology available to all to automate routine tasks. Consider this: In the 1990s, a computer cost \$20 million and occupied a whole air-conditioned room. Today, a computer with the same processing power costs \$200 in a local store and takes as much space as an average book. This dramatic change in the power and the cost of computing technology is the primary driver of Artificial Intelligence, and it is not about to go away. In many cases, through pure randomization, Artificial Intelligence is able to dramatically innovate. Consider ChatGPT and other related models known as Transformers. One of the features of transformers is literally a random number generator that "dreams up" things. This feature opens up frontiers that people literally did not have a chance to dream about. Think about it as having a creative team partner that comes up with often off-the-wall, impossibly creative ideas. Who wouldn't want to have someone like that on their team, especially when there are few expenses involved in maintaining that relationship? Now imagine an AI version of a teacher assistant available in every classroom. Someone (or something, if you want to be technical about pronouns) who always cares, keeps track of the students' work and suggests areas for improvement and challenge questions. Someone who also lives on many students' favorite objects: their phones, and nudges them to answer a random practice question or two in the middle of their TikTok session. Someone who reminds them to do their homework, praises them for their accomplishments and just generally cares. Can you imagine what kind of impact this technology may bring? As someone who has built apps and AI-based platforms for the past 20 years, I am very keen on seeing these ideas shine. The time has come to improve the U.S. education without keeping the brightest kids behind in the name of equity. Let the best kids do their best by equipping them with unlimited AI-based resources on their devices. I believe that this administration can make it happen! (And I will personally be thrilled to participate.) Many thanks in advance for your consideration. With Kind Regards, Irene Aldridge

Use AI to Defend U.S. Data In the summer of 2023, I had the privilege to consult with a major quasi-government agency entrusted with a lot of confidential information. The agency plays a critical role in American commerce, handling sensitive and detailed data. To my surprise, I discovered that all this data, including very precise identifiers, was being sent to India, where it is accessed and processed by foreign nationals beyond the jurisdiction of the United States. The implications of outsourcing data work to foreign entities should not be overlooked. Although globalization has brought many benefits, we must exercise caution when sharing sensitive data with foreign countries, particularly when outside the bounds of our legal jurisdiction. This is not a call for isolationism, but rather a plea for a thoughtful reconsideration of the risks involved. Unfortunately, the outsourcing of sensitive US data is not an isolated incident. In previous years, I happened to learn that at least some local police departments in the Northeast were shipping data, like license plate numbers, to be processed in Singapore. This accumulation of personal information in a foreign country poses a potential national security threat. We must ask why such data, which include identifiable locations, are being sent overseas without proper government clearances or protocols. Perhaps it is time to harness Artificial Intelligence to defend our data. For example, AI can be used to randomly sample the data flow from the US to foreign destinations to recognize that sensitive information is not being shipped overseas. Irene Aldridge is a quantitative researcher, CEO of AbleMarkets, coauthor of "Big Data Science in Finance" (with Marco Avellaneda) and "Real-time Risk: What Investors Should Know about Fintech, High-Frequency Trading and Flash Crashes" (with Steve Krawciw), and Adjunct Professor in the Cornell Financial Engineering program in Manhattan.